GENERALISATIONS LIFT ALONG THE SUPPORT OF A FINITE FLAT MODULE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Flatness as preservation of exactness. Used in the counted proof (lines 10080-10095). Keeps the exact sequences exact after tensoring, which is what produces the required prime.
- Support of a finite module over the base. Used to translate the topological statement (lines 10076-10080). Identifies the primes in the support with the nonvanishing localisations.

Source reference. Original lemma 00DV: sources/stacks-algebra/pinned-original-source.tex, source lines 3833–3833. Navigation pointer for \ref{lemma-NAK} used inside the counted span (line 10086); the argument that consumes it is retained in the span.

Source reference. Original lemma 00HP: sources/stacks-algebra/pinned-original-source.tex, source lines 9341–9341. Navigation pointer for \ref{lemma-ff} used inside the counted span (line 10088); the argument that consumes it is retained in the span.

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Source reference. Original lemma 00HT: sources/stacks-algebra/pinned-original-source.tex, source lines 9413–9413. Navigation pointer for \ref{lemma-flat-localization} used inside the counted span (line 10082); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $R \to S$ be a ring map. Let N be a finite S-module flat over R. Endow $\mathit{Supp}(N) \subset \mathrm{Spec}(S)$ with the induced topology. Then generalizations lift along $\mathit{Supp}(N) \to \mathrm{Spec}(R)$.*

Proof. The meaning of the statement is as follows. Let $\mathfrak{p} \subset \mathfrak{p}' \subset R$ be primes. Let $\mathfrak{q}' \subset S$ be a prime $\mathfrak{q}' \in \mathrm{Supp}(N)$ Then there exists a prime $\mathfrak{q} \subset \mathfrak{q}'$, $\mathfrak{q} \in \mathrm{Supp}(N)$ lying over $\mathfrak{p}$. As N is flat over R we see that $N_{\mathfrak{q}'}$ is flat over $R_{\mathfrak{p}'}$, see Lemma 00HT. As $N_{\mathfrak{q}'}$ is finite over $S_{\mathfrak{q}'}$ and not zero since $\mathfrak{q}' \in \mathrm{Supp}(N)$ we see that $N_{\mathfrak{q}'} \otimes_{S_{\mathfrak{q}'}} \kappa(\mathfrak{q}')$ is nonzero by Nakayama's Lemma 00DV. Thus $N_{\mathfrak{q}'} \otimes_{R_{\mathfrak{p}'}} \kappa(\mathfrak{p}')$ is also not zero. We conclude from Lemma 00HP that $N_{\mathfrak{q}'} \otimes_{R_{\mathfrak{p}'}} \kappa(\mathfrak{p})$ is nonzero. Let $J \subset S_{\mathfrak{q}'} \otimes_{R_{\mathfrak{p}'}} \kappa(\mathfrak{p})$ be the annihilator of the finite nonzero module $N_{\mathfrak{q}'} \otimes_{R_{\mathfrak{p}'}} \kappa(\mathfrak{p})$. Since J is a proper ideal we can choose a prime $\mathfrak{q} \subset S$ which corresponds to a prime of $S_{\mathfrak{q}'} \otimes_{R_{\mathfrak{p}'}} \kappa(\mathfrak{p})/J$. This prime is in the support of N, lies over $\mathfrak{p}$, and is contained in $\mathfrak{q}'$ as desired. □